21.FIND-S Algorithm

Aim

To implement the FIND-S algorithm to find the most specific hypothesis consistent with a given set of training examples stored in a CSV file.

Algorithm + Procedure

1. Start with the most specific hypothesis, represented by 0 for all attributes.
2. Read the training examples from the CSV file.
3. Check each training example one by one.
4. If the example is positive (Yes), compare it with the current hypothesis.
5. If an attribute is 0, replace it with the corresponding value from the positive example.
6. If the attribute value differs from the current hypothesis, replace it with ?.
7. Ignore all negative examples.
8. Repeat the process for all training examples.
9. Display the final hypothesis.

Code

```
import csv

# Create EnjoySport dataset

data = [

    ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'EnjoySport'],

    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],

    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']

]

# Create CSV file

with open('enjoysport.csv', 'w', newline='') as f:

    writer = csv.writer(f)

    writer.writerows(data)
```

```
# FIND-S Algorithm
def find_s(filename):
   with open(filename) as f:
      data = list(csv.reader(f))
   hypothesis = ['0'] * (len(data[0]) - 1)
   for row in data[1:]:
      if row[-1].strip().lower() == 'yes':
         for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
               hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
               hypothesis[i] = '?'
   return hypothesis
print("Most specific hypothesis:", find_s('enjoysport.csv'))
```

Output:

```
Most specific hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Result

The FIND-S algorithm was successfully implemented and the most specific hypothesis consistent with the positive training examples was obtained.